

Jonathan Prokos

Security Engineer & Researcher | ML-Powered Threat Detection | Vulnerability Analysis | (D)ARPA PI
Arlington, VA | (561) 316-6908 (VoIP) | [jobs AT prokos.us](#) | [jonathan.prokos.us](#) | Active TS

WORK HISTORY

Two Six Technologies

Arlington, VA

Lead Security Research Engineer

October 2025 - Present

- Lead adversarial security research programs spanning ML systems and binary reverse engineering
 - PI on DARPA [COOP](#), lead IV&V team conducting hardware fault modeling & evaluation
 - Integration lead on graph-based program, unifying research into cohesive T&E platform
- Built SMT-based ghidra extension to find multi-step backdoors in FOSS 10-100x faster than existing tooling. Demo'd and secured funding for future development into full toolkit
- Lead writer on \$20M+ proposals; identify and translate operational gaps into fundable research

Senior Security Research Engineer

Jan 2024 - September 2025

- Led first phase of [HARDEN](#) program as Principal Investigator; secured continuation of work through downselection; recovered healthy profit margin from poor financial position
 - Built automated, scalable *vulnerability-assessment pipeline* for UEFI/Linux kernel “[weird machines](#)” chaining exploit primitives to synthesize complete
- Led 3 direct reports on 5G/NR wireless program, presented biweekly findings to team & customer
- Led technical direction of DARPA proposal focused on deriving demographic info from code
 - Led discussions with internal & external partners to develop finalized text & figures
- Interfaced directly with customers to interpret high level requirements and provide technical insight
- Delivered high-priority demos and technical reports under compressed timelines across programs

Staff Security Research Engineer

July 2022 - Dec 2023

- SME on red/blue AI program; built adversarial testing platform to evaluate defensive robustness
 - Developed multimodal attacks to bypass SOA defenses with constrained threat models
- Developed binary similarity analysis framework utilizing elasticsearch database for efficient search
 - Published [blog post](#) outlining development process to bolster company brand
- Developed dataflow analysis for COTS vulnerability detection platform using Neo4j & BinaryNinja
- Built ML data pipeline for audio-physical side-channel analysis project; packaged final deliverable

Matthew Green Lab - Johns Hopkins Information Security Institute

Baltimore, MD

Research Assistant - First author and [presenter](#) for publication in [USENIX '23](#)

June 2020 – June 2022

- Performed security analysis of locality sensitive image hashing algorithms (PhotoDNA, PDQ) to assess resistance to adaptive privacy attacks using novel ML techniques
- Designed formal threat models of adaptive attack vectors in deployed cryptographic systems
- Developed ML pipeline for testing new hashing algorithms against these threat models

EDUCATION

Johns Hopkins University

Baltimore, MD

Bachelor of Science in Computer Science, Applied Math & Statistics Minor

May 2021

Master of Science in Security Informatics - [Masson Fellowship](#)

May 2022

TECHNICAL SKILLS

Languages: Python, C, C++, Java, Bash, HTML/CSS, JQuery, MatLab, R, Golang, x86, Rust

ML / AI: PyTorch, TensorFlow, adversarial ML, agentic AI systems, MLOps, Slurm

Systems & Security: BinaryNinja, Ghidra, IDA Pro, QEMU, Kali, UEFI/kernel exploitation, 5G/NR

DevOps & Infra: Docker, Git, CI/CD, ElasticSearch, Neo4j, Linux/Unix, T&E platform development

Other: Proposal writing, customer engagement, Agile/Scrum, program management, people management

Research Interests: Adversarial AI, autonomous systems security, threat modeling, binary RE, privacy

Recommendations may be furnished upon request.